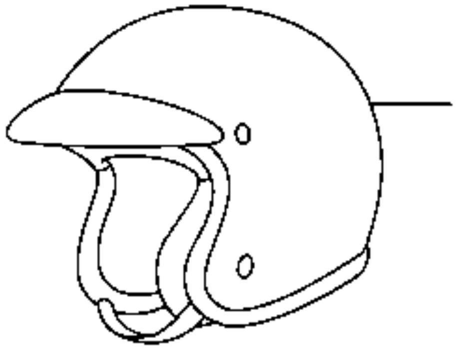
Bahagian A

[60 markah]

Jawab **semua** soalan dalam bahagian ini.

1 **Rajah 1 menunjukkan topi keledar yang diperbuat daripada bahan komposit Q.**
Diagram 1 shows a helmet that is made of composite material Q.



Rajah 1
Diagram 1

(a) Nyatakan maksud bahan komposit.
State the meaning of composite material.

(b) Kenal pasti bahan matriks dan bahan pengukuhan dalam topi keledar.
Identify the matrix substance and the strengthening substance in the helmet.

(i) Bahan matriks : ....
Matrix substance

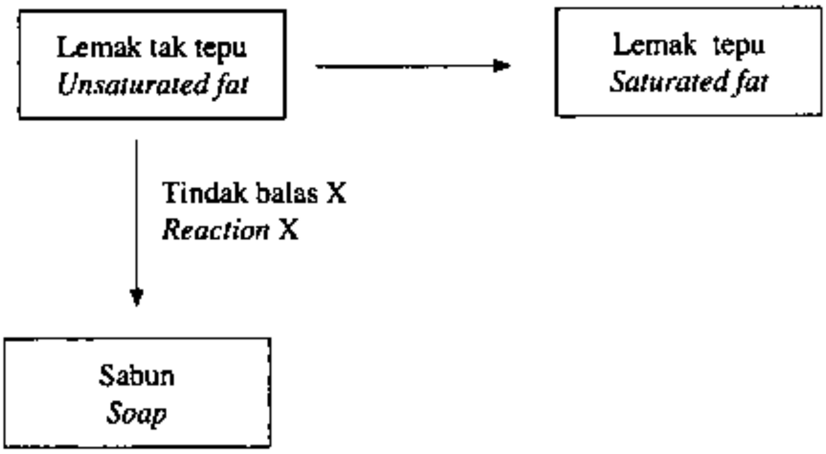
(ii) Bahan pengukuhan : ....
Strengthening substance

(c) Nyatakan **dua** sifat bahan komposit Q yang membolehkannya sesuai digunakan untuk membuat topi keledar itu.

State **two** *properties of composite material Q that make it suitable to be used to make the helmet.*

2 Rajah 2 menunjukkan tindak balas yang melibatkan lemak tak tepu.

Diagram 2 shows the reactions that involved unsaturated fat.



Rajah 2
Diagram 2

(*a*) Nyatakan **satu** contoh lemak tak tepu.

State **one** *example of unsaturated fat.*

(*b*) (i) Nyatakan nama tindak balas untuk menukarkan lemak tak tepu kepada lemak tepu.

State the name of the reaction to convert unsaturated fat to saturated fat.

(ii) Apakah kesan tindak balas dalam **2**(*b*)(i) ke atas keadaan fizik lemak?

What is the effect of the reaction in **2**(*b*)(i) *on the physical state of fat?*

(iii) Berdasarkan Rajah 2, nyatakan nama bagi tindak balas X.

Based on Diagram 2, *state the name of reaction* X.

(*c*) Tokoferol ditambah ke dalam lemak tak tepu untuk mengelakkannya menjadi tengik apabila terdedah kepada udara.

Nyatakan jenis bahan tambah makanan bagi tokoferol.

Tocopherol is added into unsaturated fat to prevent it from turning rancid when exposed to air.

State the type of food additive for tocopherol.

3 Jadual 1 menunjukkan maklumat tentang isotop magnesium

Table 1 shows an information about isotopes of magnesium.

Isotop ***Isotope***	**Kelimpahan semula jadi** ***Natural abundance***	**Jisim isotop** ***Mass of isotope***	**Nombor proton** ***Proton number***
Magnesium-24	79.0%	24	12
Magnesium-25	10.0%	25	12
Magnesium-X	11.0%	X	12

Jadual 1

Table 1

(*a*) Nyatakan maksud nombor proton.

State the meaning of proton number.

(*b*) Mengapakah isotop magnesium-24 dan isotop magnesium-25 mempunyai jisim isotop yang berlainan?

Why do the magnesium-24 and magnesium-25 isotopes have different masses of isotope?

(*c*) Lukis struktur atom bagi isotop magnesium-25.

Draw the atomic structure for magnesium-25 isotope.

(*d*) Hitung jisim isotop bagi magnesium-X.

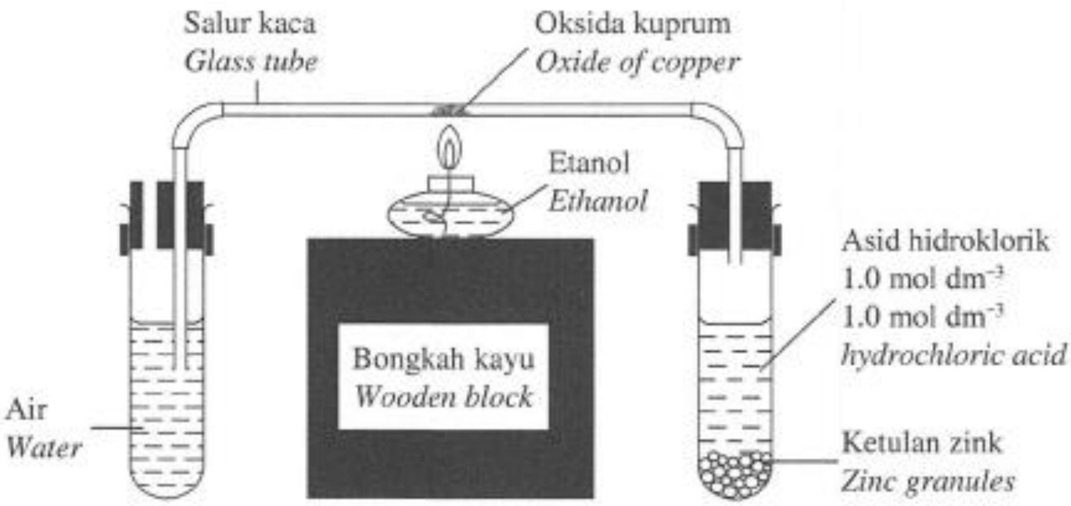
[Jisim atom relatif magnesium = 24.32]

Calculate the mass of isotope for magnesium-X.

[*Relative atomic mass of magnesium* = 24.32]

4 Rajah 3 menunjukkan susunan radas bagi satu eksperimen untuk menentukan formula empirik bagi oksida kuprum.

Diagram 3 shows the apparatus set-up for an experiment to determine the empirical formula of oxide of copper.



Rajah 3
Diagram 3

Jadual 2 menunjukkan keputusan bagi eksperimen tersebut.

Table 2 shows the results of the experiment.

Perkara ***Description***	**Jisim (g)** ***Mass* (g)**
Salur kaca *Glass tube*	4.128
Salur kaca + oksida kuprum *Glass tube + oxide of copper*	4.318
Salur kaca + kuprum *Glass tube + copper*	4.280

Jadual 2
Table 2

(*a*) Apakah yang dimaksudkan dengan formula empirik?

What is meant by empirical formula?

(*b*) Gas hidrogen dialirkan selama 10 saat sebelum pemanasan dimulakan. Aliran gas hidrogen diteruskan sehingga salur kaca disejukkan kepada suhu bilik setelah pemanasan dihentikan.

Jelaskan mengapa.

Hydrogen gas is flowed for 10 *seconds before heating is started. The flow of hydrogen gas is continued until the glass tube is cooled to room temperature after heating is stopped.*

Explain why.

(*c*) Berdasarkan keputusan eksperimen dalam Jadual 2, tentukan formula empirik bagi oksida kuprum.

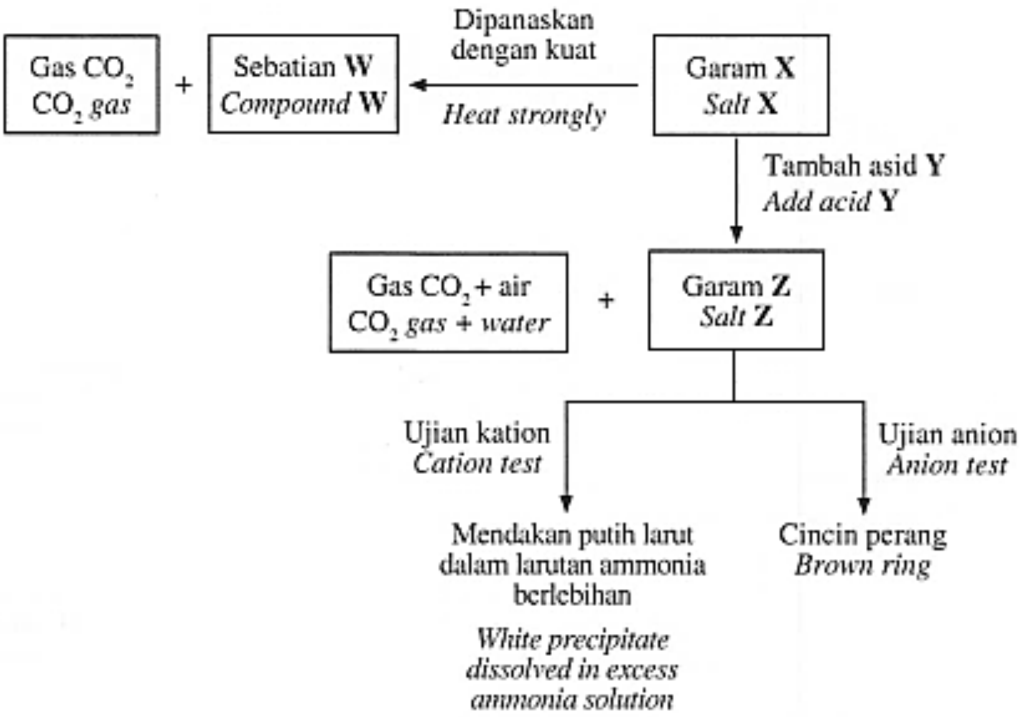
Based on the results in Table 2, *determine the empirical formula of oxide of copper.*

[Jisim atom relatif: Cu = 64 ; O = 16]

[*Relative atomic mass*: Cu = 64 ; O = 16]

5 Rajah 4 menunjukkan satu siri tindak balas bagi garam **X**.

Diagram 4 *shows a series of reactions for salt* **X**.



Rajah 4
Diagram 4

(*a*) Garam **X** adalah garam tak terlarutkan.

Nyatakan nama bagi tindak balas untuk menghasilkan garam tak terlarutkan.

Salt **X** *is an insoluble salt.*

State the name of the reaction to produce insoluble salt.

(*b*) Garam **X** dipanaskan dengan kuat untuk menghasilkan sebatian **W** dan gas karbon dioksida. Sebatian **W** berwarna kuning semasa panas dan putih semasa sejuk.

Salt **X** *is heated strongly to produce substance* **W** *and carbon dioxide gas. Compound* **W** *is yellow when hot and white when cold.*

(i) Nyatakan nama bagi sebatian **W**.

State the name of compound **W**.

(ii) Tulis persamaan kimia bagi pemanasan garam **X**.

Write the chemical equation for the heating of salt **X**.

(iii) Hitung isi padu gas karbon dioksida yang terhasil pada keadaan bilik apabila 10 g garam **X** dipanaskan.

[Jisim formula relatif **X** = 125 g mol^{-1}, 1 mol gas menempati 24 dm^3 pada keadaan bilik]

Calculate the volume of carbon dioxide gas produced at room condition when 10 g *of salt* **X** *is heated.*

[*Relative formula mass of* **X** = 125 g mol^{-1}, 1 mol *of gas occupies* 24 dm^3 *at room condition*]

(*c*) Garam **X** bertindak balas dengan asid **Y** untuk menghasilkan garam **Z**, gas karbon dioksida dan air.

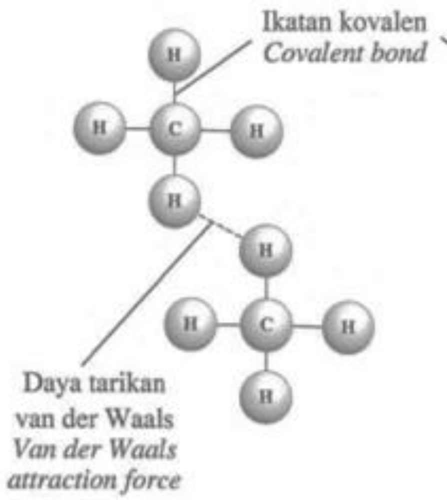
Kenal pasti asid **Y** dan garam **Z**.

Salt **X** *reacts with acid* **Y** *to produce salt* **Z**, *carbon dioxide gas and water.*

Identify acid **Y** *and salt* **Z**.

6 Jadual 3 menunjukkan dua jenis struktur molekul bagi sebatian kovalen iaitu struktur **P** dan struktur molekul gergasi.

Table 3 shows two types of molecular structures for covalent compounds which are structure **P** *and giant molecular structure.*

Struktur molekul *Molecular structure*	**Struktur P** *Structure P*	**Struktur molekul gergasi** *Giant molecular structure*
Rajah ikatan *Diagram of bonding*		

Jadual 3
Table 3

(*a*) Nyatakan **satu** contoh sebatian kovalen dengan struktur molekul gergasi.

State **one** *example of a covalent compound with giant molecular structure.*

(*b*) Berdasarkan Jadual 3,

Based on Table 3,

(i) kenal pasti jenis struktur **P**

identify the type of structure **P**

(ii) banding takat lebur dan takat didih bagi bahan yang mengandungi struktur **P** dan bahan yang mengandungi struktur molekul gergasi.

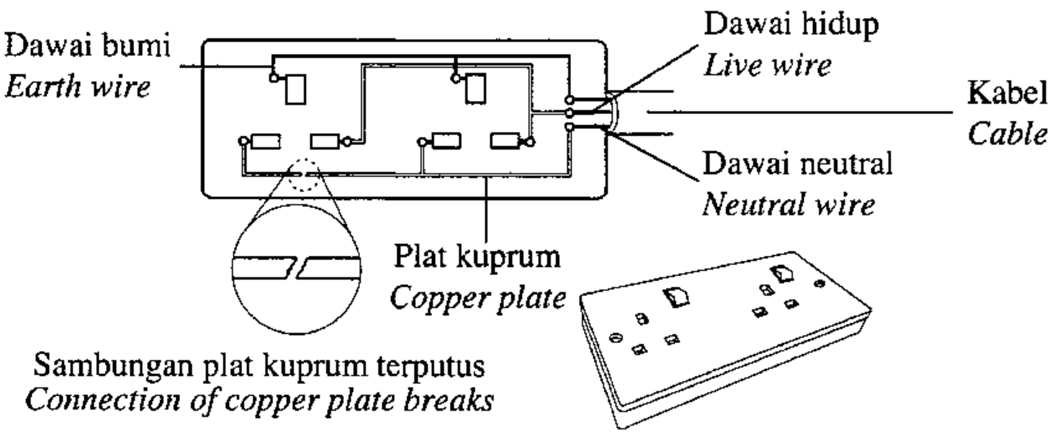
Terangkan jawapan anda.

compare the melting and boiling points of substances with structure **P** *and substances with giant molecular structure.*

Explain your answer.

(*c*) Rajah 5 menunjukkan sambungan plat kuprum yang terputus pada litar elektrik dalam satu soket berkembar yang rosak.

Diagram 5 shows a cut-off connection of a copper plate on an electric circuit in a broken double socket.



Rajah 5
Diagram 5

(i) Bagaimanakah anda dapat memperbaiki soket berkembar yang rosak itu?

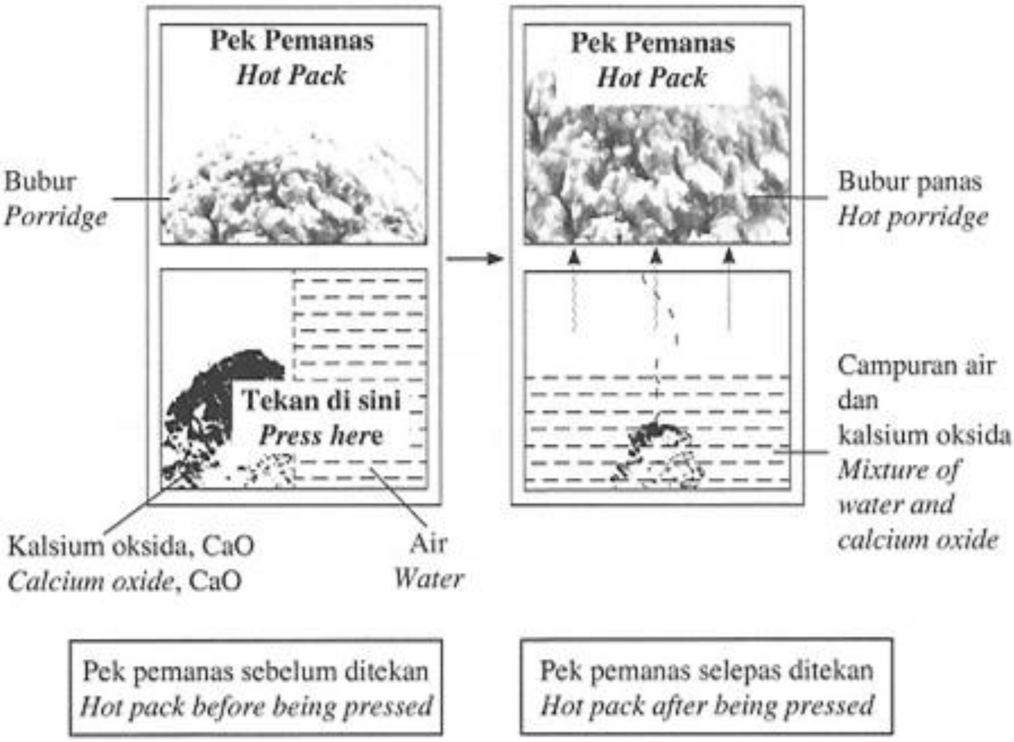
How can you repair the broken double socket?

(ii) Dengan menggunakan konsep ikatan logam, huraikan bagaimana tenaga elektrik dapat dialirkan semula setelah pengubahsuaian dilakukan.

By using the concept of metallic bond, describe how electrical energy can be flowed again after the modification has been done.

7 Rajah 6 menunjukkan tindakan yang dilakukan oleh Aina untuk memanaskan bubur dalam pek pemanas kendiri.

Diagram 6 shows the action taken by Aina to heat the porridge in a self-heating pack.



Rajah 6
Diagram 6

(*a*) (i) Nyatakan jenis tindak balas yang terlibat yang menyebabkan suhu air meningkat dalam pek pemanas kendiri.

State the type of reaction involved that causes the water temperature to rise in the self-heating pack.

(ii) Bandingkan jumlah kandungan tenaga hasil tindak balas dengan jumlah kandungan tenaga bahan tindak balas berdasarkan jawapan di 7(*a*)(i).

Compare the total energy content of the product with the total energy content of reactants based on the answer in 7(*a*)(i).

(iii) Berikut adalah persamaan termokimia bagi tindak balas antara air dengan kalsium oksida yang berlaku di dalam pek pemanas kendiri tersebut.

The following is the thermochemical equation for the reaction between water and calcium oxide that occured in the self-heating pack.

$$CaO + H_2O \longrightarrow Ca(OH)_2 \qquad \Delta H = -64 \text{ kJ mol}^{-1}$$

Tentukan perubahan suhu apabila 200 cm^3 air bercampur dengan 33.6 g kalsium oksida dalam bekas plastik.

Determine the temperature change when 200 cm^3 *of water is mixed with* 33.6 g *of calcium oxide in a plastic container.*

[Muatan haba tentu larutan, $C = 4.2$ Jg^{-1} $°C^{-1}$][Ketumpatan air = 1 g cm^{-3}]

[Jisim atom relatif: Ca = 40 , O =16]

[*Specific heat capacity of solution*, $C = 4.2$ Jg^{-1} $°C^{-1}$]

[*Density of water* = 1 g cm^{-3}] [*Relative atomic mass*: Ca = 40 , O =16]

(*b*) Jadual 4.1 menunjukkan haba peneutralan apabila dua jenis asid monoprotik, A dan B, bertindak balas dengan larutan kalium hidroksida.

Table 4.1 *shows the heat of neutralisation when two types of monoprotic acids*, A *and* B, *react with potassium hydroxide solution.*

Jenis asid ***Type of acid***	**A**	**B**
Haba peneutralan (kJ mol^{-1}) ***Heat of neutralisation* (kJ mol^{-1})**	- 55	- 57

Jadual 4.1
Table 4.1

Terangkan mengapa terdapat perbezaan nilai haba peneutralan bagi asid A dan asid B.

Explain why there is a difference in the value of heat of neutralisation for acids A *and* B.

(*c*) Jadual 4.2 menunjukkan nilai bagi dua jenis bahan api.

Table 4.2 *shows the values for two types of fuels.*

Bahan api ***Fuel***	Petrol *Petrol*	Hidrogen *Hydrogen*
Nilai bahan api (kJ g^{-1}) ***Fuel value* (kJ g^{-1})**	34	143

Jadual 4.2
Table 4.2

Berdasarkan Jadual 4.2, pilih bahan api yang lebih sesuai digunakan dalam kenderaan.

Wajarkan jawapan anda.

Based on Table 4.2, *choose a more suitable fuel to be used in a vehicle.*

Justify your answer.

8 (*a*) Jadual 5 menunjukkan bahan tindak balas dan suhu yang digunakan dalam Eksperimen I dan Eksperimen II untuk mengkaji kesan suhu ke atas kadar tindak balas.

Table 5 shows the reactants and the temperatures used in Experiment I *and Experiment* II *to study the effect of temperature on the rate of reaction.*

Eksperimen *Experiment*	**Bahan tindak balas** *Reactant*	**Suhu (°C)** *Temperature* (°C)
I	2.4 g pita magnesium + 50 cm^3 asid sulfurik 2.0 mol dm^{-3} 2.4 g *magnesium ribbon* + 50 cm^3 of 2.0 mol dm^{-3} *sulphuric acid*	30.0
II	2.4 g pita magnesium + 50 cm^3 asid sulfurik 2.0 mol dm^{-3} 2.4 g *magnesium ribbon* + 50 cm^3 of 2.0 mol dm^{-3} *sulphuric acid*	45.0

Jadual 5
Table 5

(i) Nyatakan maksud kadar tindak balas.

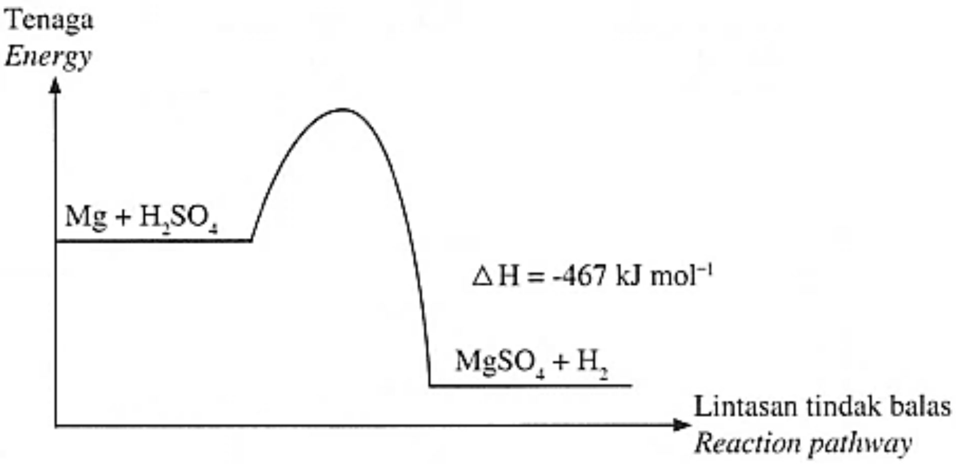
State the meaning of rate of reaction.

(ii) Berdasarkan Jadual 5, lakarkan graf isi padu gas terbebas melawan masa untuk Eksperimen I dan Eksperimen II pada paksi yang sama.

Based on Table 5, sketch a graph of the volume of gas released against time for Experiment I *and Experiment* II *on the same axes.*

(iii) Tindak balas antara magnesium dan asid sulfurik adalah tindak balas eksotermik. Rajah 7 menunjukkan gambar rajah profil tenaga bagi tindak balas itu.

The reaction between magnesium and sulphuric acid is an exothermic reaction. Diagram 7 shows the energy profile diagram for the reaction.



Rajah 7
Diagram 7

Tandakan tenaga pengaktifan dengan menggunakan simbol E_a pada Rajah 7.

Mark the activation energy by using symbol E_a *on Diagram* 7.

(iv) Hitung isi padu gas hidrogen yang terbebas dalam Eksperimen II.

Calculate the volume of hydrogen gas released in Experiment II.

[Jisim atom relatif: Mg = 24]

[Isi padu molar gas pada keadaan bilik: 24 $dm^3\ mol^{-1}$]

[*Relative atomic mass*: Mg = 24]

[*Molar volume of gas at room conditions*: 24 $dm^3\ mol^{-1}$]

(*b*) Pn. Aminah mempunyai pilihan sama ada untuk menggoreng atau merebus ubi kayu untuk menjamu tetamunya tetapi beliau mengambil keputusan untuk menggoreng ubi kayu itu.

Dengan menggunakan konsep kadar tindak balas yang anda pelajari, wajarkan cara memasak yang dipilih oleh Pn. Aminah.

Pn. Aminah has a choice of either frying or boiling some tapiocas to serve her guests but she decided to fry the tapiocas.

By using the concept of rate of reaction that you have studied, justify the cooking method chosen by Pn. Aminah.

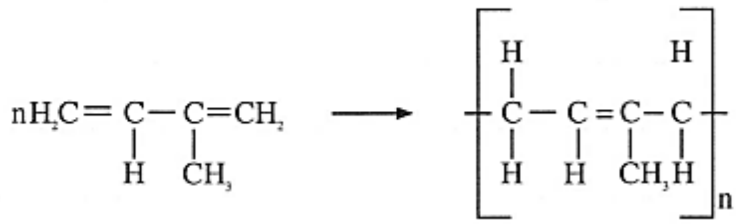
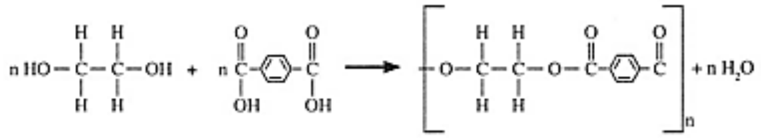
Bahagian B

[20 markah]

Jawab mana-mana **satu** soalan dalam bahagian ini.

9 (*a*) Jadual 6 menunjukkan dua jenis tindak balas pempolimeran, Tindak balas I dan Tindak balas II.

Table 6 shows two types of polymerisation reactions, Reactions I *and* II.

Tindak balas I ***Reaction* I**	
Tindak balas II ***Reaction* II**	

Jadual 6
Table 6

Berdasarkan Jadual 6, nyatakan jenis tindak balas pempolimeran I dan II.

Bandingkan kedua-dua tindak balas itu. [4 *markah*]

Based on Table 6, *state the type of polymerisation reactions* I *and* II.

Compare the two reactions. [4 *marks*]

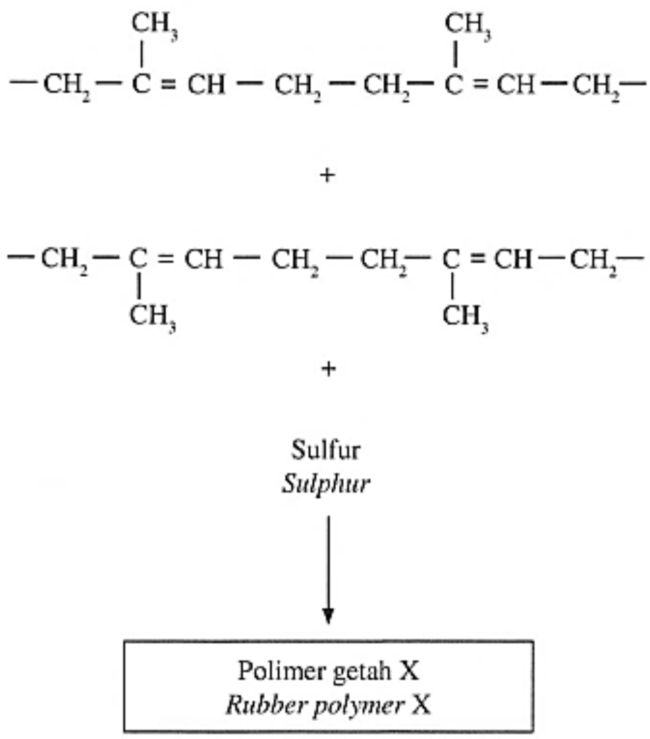
(*b*) (i) Pelupusan tayar getah terpakai secara tidak lestari akan menyebabkan pencemaran terhadap alam sekitar.

Nyatakan **dua** kaedah dan terangkan bagaimana kaedah itu dapat mengurangkan masalah tersebut. [4 *markah*]

Unsustainable disposal of used tyre will cause pollution to the environment.

State **two** *methods and explain how the methods can reduce the problem.*

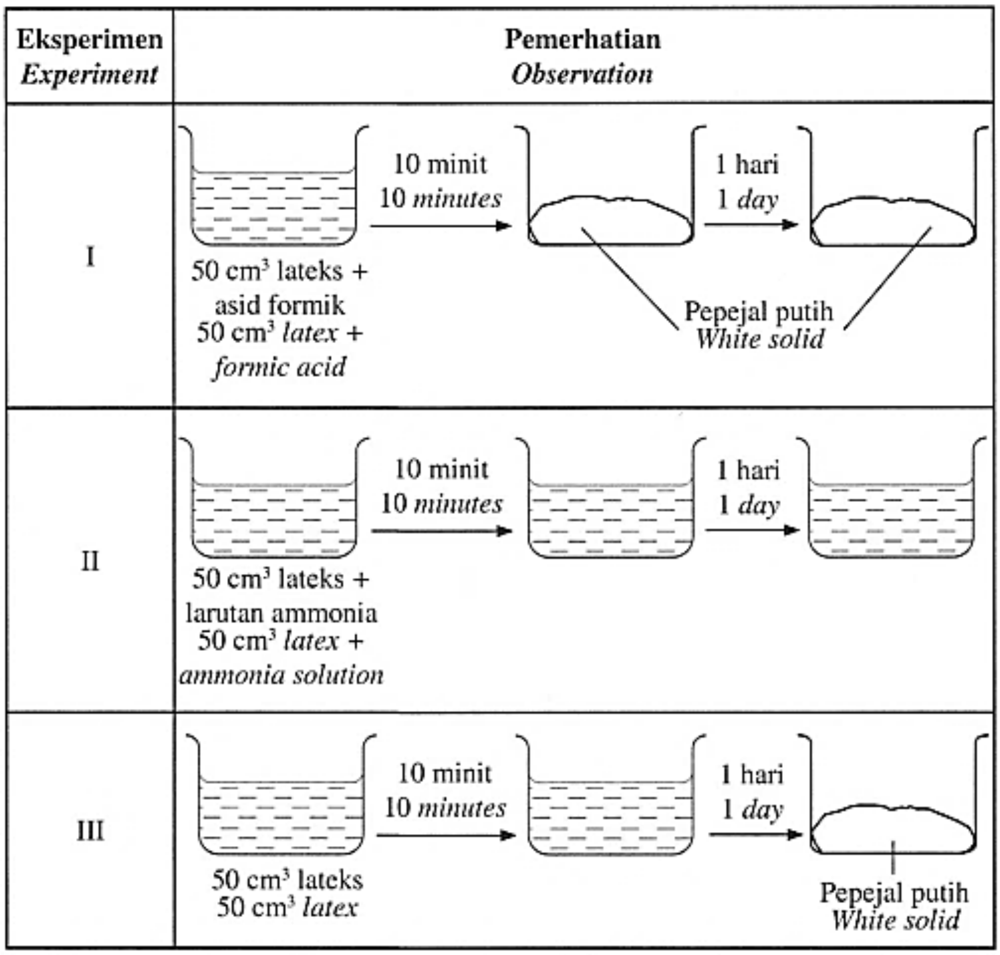
(ii) Rajah 8.1 menunjukkan suatu proses untuk menghasilkan polimer getah X.
Diagram 8.1 shows a process to produce rubber polymer X.



Rajah 8.1
Diagram 8.1

Lukis struktur polimer getah X yang terbentuk.
Draw the structure of rubber polymer X formed.

(*c*) Rajah 8.2 menunjukkan pemerhatian ke atas lateks dalam tiga eksperimen.
Diagram 8.2 shows the observations on latex in three experiments.



Rajah 8.2
Diagram 8.2

Berdasarkan maklumat dalam Rajah 8.2, terangkan perbezaan pemerhatian antara:

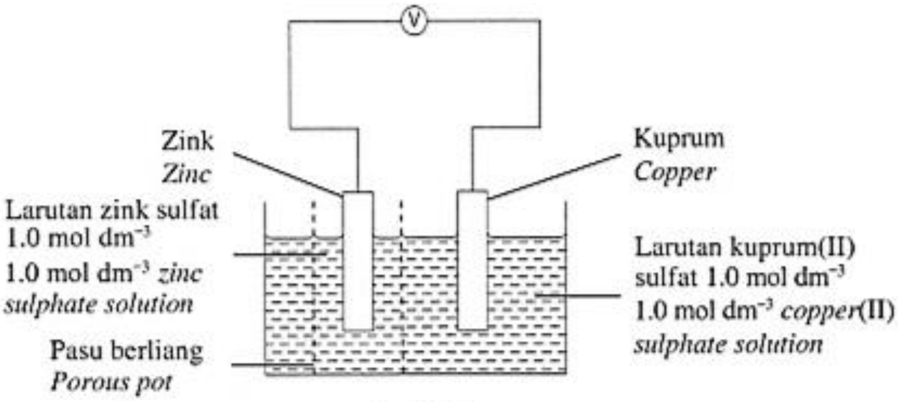
- Eksperimen I dan Eksperimen II
- Eksperimen I dan Eksperimen III

Based on the information in Diagram 8.2, explain the differences in observation between:

- *Experiment I and Experiment II*
- *Experiment I and Experiment III*

10 (*a*) Rajah 9.1 menunjukkan susunan radas bagi satu sel kimia.

Diagram 9.1 shows the apparatus set-up for a chemical cell.



Rajah 9.1
Diagram 9.1

(i) Berdasarkan Rajah 9.1, nyatakan warna larutan kuprum(II) sulfat dan fungsi pasu berliang. [2 *markah*]

Based on Diagram 9.1, state the colour of the copper(II) *sulphate solution and the function of porous pot.* [2 *marks*]

(ii) Jadual 7.1 menunjukkan nilai keupayaan elektrod piawai sel setengah bagi zink dan kuprum.

Table 7.1 shows the values of standard electrode potential for the half-cells of zinc and copper.

Tindak balas sel setengah *Reaction of half-cell*	**E^0 (V) (298 K)**
$Zn^{2+} + 2e^- \rightleftharpoons Zn$	- 0.76
$Cu^{2+} + 2e^- \rightleftharpoons Cu$	+ 0.34

Jadual 7.1
Table 7.1

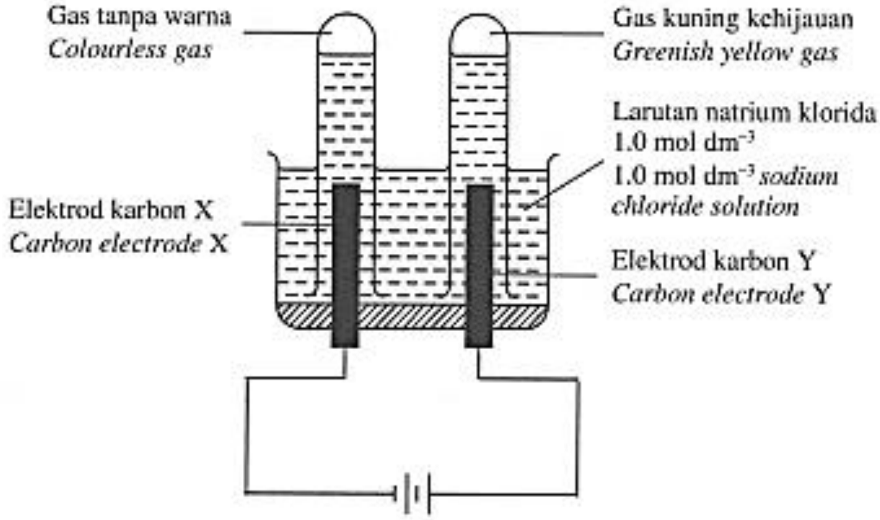
Berdasarkan Jadual 7.1, nyatakan terminal negatif dan terminal positif bagi sel itu. Tulis persamaan ion keseluruhan bagi tindak balas yang berlaku. Tulis notasi sel dan hitung nilai E^0_{sel} bagi sel kimia itu.

Based on Table 7.1, state the negative terminal and positive terminal for the cell. Write the overall ionic equation for the reaction that takes place.

Write the cell notation and calculate the E^0_{cell} value for the chemical cell.

(*b*) Rajah 9.2 menunjukkan proses elektrolisis bagi larutan natrium klorida, NaCl dengan menggunakan elektrod karbon.

Diagram 9.2 shows the electrolysis process of sodium chloride solution, NaCl *by using carbon electrodes.*



Rajah 9.2
Diagram 9.2

Jadual 7.2 menunjukkan nilai keupayaan elektrod piawai sel setengah bagi beberapa bahan.

Diagram 7.2 shows the values of standard electrode potential for the half-cells of some substances.

Tindak balas sel setengah ***Reaction of half-cell***	**E^θ (V)** **(298 K)**
$Na^+ + e^- \rightleftharpoons Na$	- 2.71
$O_2 + 2H_2O + 4e^- \rightleftharpoons 4OH^-$	+ 0.40
$2H^+ + 2e^- \rightleftharpoons H_2$	0.00
$Cl_2 + 2e^- \rightleftharpoons 2Cl^-$	+ 1.36

Jadual 7.2
Table 7.2

Berdasarkan Rajah 9.2 dan Jadual 7.2, terangkan tindak balas di elektrod X dan elektrod Y berdasarkan aspek berikut:

- ion yang tertarik ke setiap elektrod
- ion yang dipilih untuk dioksidakan dan diturunkan serta sebab ion itu dipilih
- setengah persamaan di setiap elektrod
- hasil yang terbentuk di setiap elektrod

Based on Diagram 9.2 and Table 7.2, explain the reactions at electrode X and electrode Y based on the following aspects:

- *ions that are attracted to each electrode*
- *ions that are selected to be oxidised and reduced and the reasons the ions are selected*
- *half equation at each electrode*
- *products formed at each electrode*

Bahagian C

[20 markah]

Jawab soalan dalam bahagian ini.

11 (*a*) Hidrokarbon dikelaskan kepada hidrokarbon tepu dan hidrokarbon tak tepu.

Nyatakan siri homolog bagi hidrokarbon tak tepu. Lukis formula struktur bagi dua isomer hidrokarbon tak tepu dengan empat atom karbon per molekul.

Hydrocarbon is classified into saturated and unsaturated hydrocarbon.

State the homologous series for unsaturated hydrocarbon. Draw the structural formula for two isomers of unsaturated hydrocarbon with four carbon atoms per molecule.

(*b*) Jadual 8 menunjukkan keputusan bagi dua eksperimen untuk membezakan heksana dan heksena.

Table 8 shows the results of two experiments to differentiate hexane and hexene.

Eksperimen *Experiment*	**Ujian** *Test*	**Pemerhatian** *Observation*	
		Heksana *Hexane*	**Heksena** *Hexene*
I	Tindak balas dengan larutan kalium manganat(VII) berasid *Reaction with acidified potassium manganate(VII) solution*	Warna ungu tidak berubah *Purple colour remains unchanged*	Warna ungu bertukar kepada tanpa warna *Purple colour turns to colourless*
II	Pembakaran *Combustion*	Nyalaan kuning berjelaga *Sooty yellow flame*	Nyalaan kuning lebih berjelaga *Sootier yellow flame*

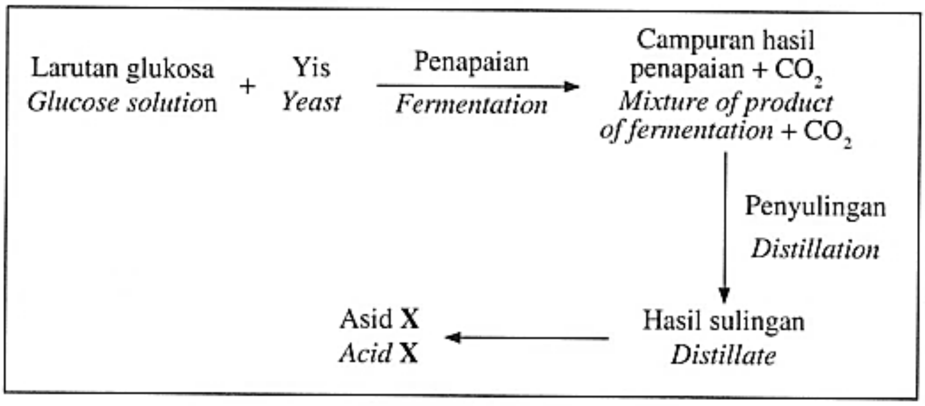
Jadual 8
Table 8

Berdasarkan Jadual 8, terangkan mengapa terdapat perbezaan pemerhatian bagi Eksperimen I. Dalam penerangan anda, tulis persamaan kimia bagi tindak balas yang berlaku. Bagi Eksperimen II, bandingkan peratus karbon per unit molekul bagi heksana dengan heksena.
[Jisim atom relatif: C = 12 ; H = 1]

Based on Table 8, explain why there is a difference in the observations for Experiment I. In your explanation, write the chemical equation for the reaction that takes place. For Experiment II, compare the percentage of carbon per unit molecule for hexane and hexene.
[*Relative atomic mass*: C = 12 ; H = 1]

(*c*) Rajah 10 menunjukkan satu carta alir bagi menghasilkan asid **X**.

Diagram 10 *shows a flow chart to produce acid* **X**.



Rajah 10
Diagram 10

Tulis persamaan kimia yang mewakili tindak balas penapaian larutan glukosa. Kenal pasti asid **X** dan huraikan bagaimana asid **X** dapat dihasilkan daripada hasil sulingan. Sertakan rajah berlabel dalam jawapan anda.

*Write the chemical equation that represents the fermentation reaction of glucose solution. Identify acid **X** and explain how acid **X** can be produced from the distillate. Include a labelled diagram in your answer.*

KERTAS PEPERIKSAAN TAMAT